a
Science
Kidney multiome-based genetic scorecard reveals convergent coding and regulatory variants
Liu H, ... Susztak K.
Science 2025
doi:10.1126/science.adp4753
Title, DOI, PMID or citation
/igvf-replicate-paper
IGVFagent
liu2025-open4gene: hurdle negative-binomial peak→gene test (port of R/Open4Gene.R; zero-model β and P identical)
encode-re2g: ENCODE-rE2G enhancer→gene model, 9 pretrained (max |Δscore| 7.8e-16 vs the authors' K562 test)
esig_net_predict: eSIG-Net inference on the released checkpoint (bit-identical to the authors' test.sh)
Deposited data: Figshare, Supp. Tables, Source Data
Authors' code at pinned commits, run as reference
Connect to an agent
IGVFagent
User: Replicate Liu et al. 2025, Science (doi:10.1126/science.adp4753)
Agent: 11 of 16 analyses reproduced.
Fig. 3D re-drawn: R = 0.520 (paper 0.51).
2 blocked: individual-level data (controlled access).
Ported methods (CLI tools)
Data
Reference code
b
/igvf-replicate-paper Liu et al. 2025 Science
1 igvfagent bench resolve → PMID 39913582 · PMC12013656
2 igvfagent bench harvest → Figshare 26299093 · 2 repos pinned
3 igvfagent bench plan --seed → 16 analyses, one per figure panel
4 igvfagent paper-code pipeline → authors' scripts = reference
5 igvfagent port register → liu2025-open4gene, liu2025-gwas-loci
6 igvfagent bench verify-port → port vs R pscl::hurdle: P identical
7 igvfagent bench run → recompute Figs. 1–6 from tables
8 igvfagent figures redraw → Fig. 3D redrawn, side by side
9 igvfagent bench score → reproduction: incomplete, 11/16
loop 4–9 until every analysis is reproduced or blocked
Chain of tools
Output: Benchmarks/liu2025_kidney_multiome/README.md
Coverage 11/16 analyses reproduced; 43/46 checks pass
Ports liu2025-open4gene, liu2025-gwas-loci, each verified against the authors' own code
Exact ASE genes 8,573 / 8,725 / 10,398; bASA peaks 19,083; Open4Gene links 142,452; coding 1,363
Differs median peaks per gene 4 vs 8 (author re-checking); non-MHC variants +2.9%
Blocked RASQUAL, snASA, CARMA: individual-level genotypes, controlled access
Asks a table of exact data needed from the authors, ordered by how much each item unlocks
Output
c
User: Replicate this paper with IGVFagent
Liu et al. 2025, Science
Fig. 3D: chromatin effect vs expression effect (R = 0.520, paper 0.51)
Effect size of Tubule ASE
Effect size of bASA
Gschwind et al. 2026, Nature
Fig. 2b,c: CRISPR benchmark AUPRC (ours from Supp. Table 3 vs paper)
paper
IGVFagent
AUPRC (CRISPR benchmark)
0.66
0.56
0.74
0.61
ENCODE-rE2G
ABC
rE2G Extended
ABC (Ext. set)
Pan et al. 2026, Nat Methods
Fig. 1c: eSIG-Net finds disease mutations (ROC-AUC 0.908, paper 0.91)
True positive rate
False positive rate
eSIG-Net
DeepFE
D-SCRIPT
PPR
PLM-interact
SDNN
19 of 20 numbers within 5% of the paper
GWAS variants, P<5e-8
non-MHC variants
independent loci
ASE genes, tubule
ASE genes, glomerulus
ASE genes, union
bASA peaks
bASA SNPs
bASA vs ASE R
snASA SNPs
snASA in open chromatin (%)
PT snASA vs bASA rho
Open4Gene links
Open4Gene peaks
median peaks/gene
GWAS variants in peaks
coding variants
coding genes
convergent genes
validated (%)
-50%
IGVFagent vs paper (%)
11 of 11 numbers within 5% of the paper
biosamples
cell types / tissues
elements per biosample
links per biosample
predicted links, total
genes per enhancer
enhancers per gene
AUPRC, rE2G
AUPRC, ABC
AUPRC, rE2G Extended
AUPRC, ABC (Extended set)
IGVFagent vs paper (%)
12 of 12 numbers within 5% of the paper
accuracy, disease
ROC-AUC, disease
AP, disease
accuracy, population
ROC-AUC, population
AP, population
ROC-AUC vs structure
AP vs structure
FoldDock mutant acc.
FoldDock consistency (%)
FoldDock 17.6% (%)
FoldDock WT acc.
IGVFagent vs paper (%)
11 of 16 analyses reproduced
11
3
2
5 of 5 analyses reproduced
5
11 of 16 analyses reproduced
11
1
4
reproduced (paper colour)
partial / not reproduced
blocked or pending